HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Executed thermal mapping studies and process development for pharmaceutical sterilisation equipment, analysing validation data to identify performance bottlenecks.
- Supported commissioning, qualification, and validation (CQV) activities for critical process systems within **GMP-regulated manufacturing environments**, ensuring adherence to quality standards.
- Developed and executed validation protocols and qualification documentation for autoclave and washer equipment, contributing to the equipment qualification lifecycle.
- Performed thermocouple calibration and verification activities, maintaining instrument accuracy for critical process monitoring and validation studies.
- Analysed process performance issues and implemented improvements, troubleshooting equipment behaviour to optimise operational efficiency.
- Liaised with **clients, suppliers, and internal technical specialists** to gather requirements and present technical findings, facilitating project progress.
- Managed multiple validation priorities concurrently, working independently to meet project timelines and deliver comprehensive technical reports.

SKILLS

Life Sciences & GMP – GMP, Pharmaceutical Manufacturing, Biopharmaceutical Manufacturing, Sterilisation Validation, Process Development, Thermal Mapping, Equipment Qualification, Validation Protocols, CQV

Engineering & Validation – Process Systems, Autoclave Qualification, Washer Qualification, Temperature Mapping, Instrument Calibration, Technical Writing, Report Generation, Troubleshooting, Quality Standards

Project & Client Management – Customer-Facing, Technical Presentations, Client Liaison, Supplier Management, Independent Work, Priority Management, Site Travel

Data Analysis & Reporting – Validation Data Analysis, Performance Studies, Technical Documentation, Interpret Technical Drawings, Interpret Specifications

Core Technologies – E-Val Pro, GDP Documentation Practices, Pharmaceutical Process Equipment

PROJECTS

Pharmaceutical Sterilisation Process Optimisation (HCL Technologies)

- Designed and executed validation studies for a new **autoclave sterilisation process**, focusing on cycle development and performance qualification for pharmaceutical product batches.
- Analysed thermal mapping data using **Excel and statistical methods** to confirm uniform temperature distribution and validate process parameters against client specifications.
- Authored detailed technical reports and qualification documentation, presenting findings and process recommendations to the client's operations and quality assurance teams.

GMP Equipment Validation Protocol Development (HCL Technologies)

- Developed comprehensive validation protocols for a new **industrial washer system**, detailing test procedures, acceptance criteria, and required documentation for GMP compliance.
- Collaborated with cross-functional teams, including client engineers and validation specialists, to ensure protocols accurately reflected system functionality and regulatory requirements.
- Supported the execution of IQ/OQ/PQ phases, including instrument calibration verification and data collection, to formally qualify the equipment for biopharmaceutical manufacturing.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)